

Trishna Saikia

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Google Scholar

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EDUCATION

- **Indian Institute of Technology** Indore, India
Doctor of Philosophy - Computer Science and Engineering ; CPI: 8.5/10 Aug 2022 - Aug 2026 (Expected)
- **National Institute of Technology** Jamshedpur, India
Master of Technology - Computer Science and Engineering; CPI: 9.18/10 Aug 2020 - Jul 2022
- **Gauhati University Institute of Science and Technology** Guwahati, India
Bachelor of Technology - Information Technology; CPI: 9.02/10 Aug 2015 - Jul 2019

RESEARCH PROFILE

PhD researcher in Computer Science and Engineering with experience in machine learning, computer vision, biomedical signal analysis, digital health, and healthcare-oriented AI. My research focuses on developing robust AI methods for physiological signal analysis, non-contact health monitoring, and biomedical data modelling under real-world conditions. I have worked on healthcare dataset collection, preprocessing, model development, evaluation, and physiological cue-based assessment, including pain analysis and neonatal pain detection from facial videos. My current research interests include AI-based digital health, physiological sensing, clinical decision support, emergency-care AI, and trustworthy machine learning for healthcare applications.

ONGOING RESEARCH

- **Machine Learning and Biomedical Signal Analysis for Digital Health Applications** July 2022 - Present
 - Developed deep learning methods for non-contact physiological monitoring from facial videos.
 - Applied AI-based physiological sensing to respiratory rate, blood pressure, SpO₂, and heart-rate estimation.
 - Worked on healthcare-related dataset collection, preprocessing, and robust model evaluation.
 - Explored pain assessment and neonatal pain detection using physiological and multimodal cues.
 - Worked with noisy real-world biomedical data affected by motion, occlusion, compression, and illumination variation.
- **Physiological Cue-Based Deepfake and Affective Computing Research** July 2022 - Present
 - Worked on pain assessment and neonatal pain detection using physiological and multimodal cues.

PATENTS

- Puneet Gupta, **Trishna Saikia**, Anup Kumar Gupta.: "System, apparatus, and method for heart rate determination of individuals wearing face mask", Under filing in Indian jurisdiction (Ref. No: 202321071568).

PUBLICATIONS

- **Trishna Saikia**, Anup Kumar Gupta, Puneet Gupta, and Pasi Liljeberg.: "Exploring PPG-guided Knowledge Distillation for Contactless Respiration Estimation", *IEEE International Conference on Automatic Face and Gesture Recognition (2026)*, (Accepted).
- **Trishna Saikia**, Anup Kumar Gupta, Puneet Gupta, and Pasi Liljeberg.: "Respiratory Rate Estimation via Cross-Modality Supervision of Lagrangian and Eulerian rPPG with Skin-Tone and Illumination Awareness", *Image and Vision Computing*, (Submitted).
- Dhamaniya, Ashutosh, Anup Kumar Gupta, **Trishna Saikia**, and Puneet Gupta.: "Exploring Remote Photoplethysmography for Neonatal Pain Detection from Facial Videos." arXiv preprint arXiv:2604.25680 (2026).
- **Trishna Saikia**, Puneet Gupta, and Pasi Liljeberg: "A Step Forward in rPPG-Based Blood Pressure Estimation: A Novel Dataset and Baseline Method", *IEEE Transactions on Consumer Electronics*, (Under review).
- Anup Kumar Gupta, **Trishna Saikia**, Puneet Gupta, and Abhinav Dhall.: "PainXtract: A Multimodal System for Multiclass Pain Classification Using Physiological Signals", *International Conference on Multimodal Interaction (2025)*, DOI: <https://doi.org/10.1145/3747327.3764792>.
- **Trishna Saikia**, Satwik Vankayalapati, Puneet Gupta, and Pasi Liljeberg: "VOLEMIA: Non-Invasive Blood Pressure Estimation Using Temporal-Spectral Convolutional Network", *Digital Signal Processing (2025)*, DOI: <https://doi.org/10.1016/j.dsp.2025.105393>
- Vaidehi Agarwal, **Trishna Saikia**, Anup Kumar Gupta, and Puneet Gupta: "SHINE: Synergizing Transformers with Contrastive Learning for Thriving rPPG-based SpO₂ Estimation", *Expert Systems with Applications*, (2025), DOI: <https://doi.org/10.1016/j.eswa.2025.129190>.
- Lokendra Birla, **Trishna Saikia**, and Puneet Gupta: "AVENUE: A Novel Deepfake Detection Method Based on Temporal Convolutional Network and rPPG Information", *ACM Transactions on Intelligent Systems and Technology (2025)*, DOI: <https://doi.org/10.1145/3702232>
- Lokendra Birla, Sneha Shukla, **Trishna Saikia**, and Puneet Gupta: "HR-TRACK: An rPPG Method for Heart rate Monitoring Using Temporal Convolution Networks." In the *International Conference on Pattern Recognition (2024)*.
- **Trishna Saikia**, Lokendra Birla, Anup Kumar Gupta, and Puneet Gupta: "HREADAI: heart rate estimation from face mask videos by consolidating eulerian and lagrangian approaches," *IEEE Transactions on Instrumentation and Measurement (2023)*, DOI: 10.1109/TIM.2023.3334359

- **Trishna Saikia**, Ranjeet Kumar, Dilip Kumar and Koushlendra Kumar Singh: “An Automatic Lung Nodule Classification System Based on Hybrid Transfer Learning Approach”, SN Computer Science (2022), DOI: <https://doi.org/10.1007/s42979-022-01167-0>
- **Trishna Saikia**, Malho Hansdah, Koushlendra Kumar Singh, and Manish Kumar Bajpai: “Classification of Lung Nodules based on Transfer Learning with K-Nearest Neighbor (KNN)”, Imaging System and Technology (2022), DOI: <https://doi.org/10.1109/IST55454.2022.9827668>
- **Trishna Saikia**, and Koushlendra Kumar Singh: “Computer-Aided leaf Disease Classification based on Logistic Regression and Transfer Learning Approach”, In International Conference on Machine Vision and Augmented Intelligence (2022), Springer Singapore
- **Trishna Saikia**, and Koushlendra Kumar Singh: “Introduction to Various Optimization Techniques”, Nature-Inspired Optimization Algorithms and Soft Computing, IET book, Chapter-1 (ISBN: 978-1-83953-517-8)

PROJECTS

- **Vehicle Tracking and Crowdedness Detection Inside the Vehicle** *Jan. 2019 - Jul. 2019*
 - Developed a real-time bus tracking and crowdedness estimation system.
 - Worked on video/image-based analysis, detection, and practical deployment-oriented system design.
 - Relevant skills: computer vision, tracking, OpenCV, machine learning, real-time sensing.
- **Smart Face Recognizing Locking System** *Sep. 2018*
 - Developed an intelligent face-recognizing locking system at Utkranti 1.0, organized at GUIST, Guwahati.
 - Framework: OpenCV and Arduino.
- **ChatBot on NorthEast tourism** *Apr. 2018*
 - Developed a chatbot on NorthEast tourism, the Data Science Hackathon organized at IIT Guwahati.
 - http://t.me/sangrakt_bot.

RESEARCH EXPERIENCE

- Developed AI and computer vision pipelines for video-based sensing and temporal signal analysis.
- Worked with spatio-temporal deep learning models: CNNs, TCNs, Transformers, and cross-modal learning frameworks.
- Experienced in face detection, ROI extraction, visual preprocessing, temporal feature extraction, and model evaluation.
- Worked with noisy real-world video data affected by motion, occlusion, compression, and illumination variation.
- Experienced in PyTorch, TensorFlow, OpenCV, signal processing, and end-to-end research implementation.
- Published research in computer vision, signal processing, multimodal AI, healthcare AI, and physiological computing.
- Familiar with Git-based code management, documentation, and collaborative research implementation.

TEACHING AND MENTORING EXPERIENCE

- **Guest Faculty under PMRF Scheme, GUIST** *Jan. 2025 – Present*
 - Taught System Programming, Web Technology, and Python Programming.
- **Guest Faculty under PMRF Scheme, SGSITS Indore** *Mar. 2023 – Dec. 2024*
 - Taught Deep Learning and Reinforcement Learning.
- **Teaching Assistant, IIT Indore** *Aug. 2022 – Present*
 - Assisted courses: Computer Vision, Pattern Recognition, Software Engineering, and Data Structures and Algorithms.
 - Mentored 5 Master’s students and 2 junior PhD scholars at the Deep Intelligence Lab, IIT Indore.

INTERNATIONAL RESEARCH EXPERIENCE

- **Visiting Research Scholar, University of Turku, Finland** *Apr. 2026 – Jun. 2026*
 - Erasmus+ funded research stay at the Department of Computing, University of Turku.
 - Worked under the supervision of Prof. Pasi Liljeberg on AI and physiological computing research.
 - Gained experience in an international and interdisciplinary Finnish research environment.

TECHNICAL SKILLS & ACHIEVEMENTS

Technical Skills:

- **Programming:** Python, MATLAB, C
- **Machine Learning:** Classification, regression, clustering, dimensionality reduction, and model evaluation
- **Deep Learning:** PyTorch, TensorFlow, CNNs, RNNs, LSTMs, Transformers, and TCN
- **Computer Vision:** OpenCV, video processing, object/face detection, ROI extraction, visual feature extraction
- **Signal and Sensor Data Processing:** filtering, FFT, PSD analysis, feature extraction, sensor/video signal analysis
- **Data Handling and Analysis:** CSV/Excel data processing, NumPy, pandas, scikit-learn
- **Tools:** LaTeX, Git, SageMath, Microsoft Excel

Achievements:

- Selected as an India RISE Fellow (2026).
- Selected for an Erasmus+ funded visiting research stay at the University of Turku, Finland, April–June 2026.
- Institute Best Technology Award, 2025, Indian Institute of Technology (IIT) Indore.
- Prime Minister Research Fellowship scholar, IIT Indore.
- Received incubation support from Assam Startup Incubation facilities during B.Tech.
- Received Ishan Uday Fellowship, in B.Tech from the Department of IT, Gauhati University (2015-19 batch)
- 1st position in B.Tech from the Department of IT, Gauhati University (GU) (2015-19 batch).

OTHER ACTIVITIES:

- Served as a peer reviewer for International Conferences like ICASSP and journals including: IEEE Transactions on Affective Computing (TAFFC), IEEE Transactions on Systems, Man, and Cybernetics: Systems, Biomedical Signal Processing and Control, Expert Systems with Applications, Pattern Recognition, Engineering Applications of Artificial Intelligence, Signal, Image and Video Processing, and SN Computer Science.
- Contributed to research documentation: project reports, and proposal writing for interdisciplinary AI and healthcare research.
- Research Mentoring: Mentored 3 Master's students and 2 junior PhD scholars at the Deep Intelligence Lab, IIT Indore.
- Teaching Assistant for: Computer Vision, Pattern Recognition, Software Engineering, and Data Structures and Algorithms at IIT Indore.

LANGUAGE SKILLS

- **English:** Professional working proficiency
- **Hindi:** Fluent
- **Assamese:** Native

REFERENCES

- **Dr. Puneet Gupta**
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Indian Institute of Technology Indore, India
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Relationship: Collaborator